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The Estimation of Battery State of Charge using Corny Network

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Abstract*: *State of charge (SOC) estimation of lithium-ion batteries has been extensively studied, and the estimation accuracy was mainly investigated through the development of various battery models and dynamic estimation algorithms. All battery models, however, contain inherent model bias due to the simplifications and assumptions, which cannot be effectively addressed through the development of various conventional and intelligent computational methods. Consequently, some existing techniques for estimating battery state of charge (SOC) have not been very accurate in predicting battery SOC characteristics, using both traditional and intelligent computation methods. There are some drawbacks to employing deep learning to estimate SOC battery, such as the use of complicated algorithms or networks, overfitting, and others. The proposed method, the Corny architecture, has a narrow layer design. This design has low computational cost and prevents overfitting. The result shows that the accuracy of the method is very high. The predicted and targeted values are almost merged in a single line. The RMSE and MAX error indexes are very low. The accuracy of the model is acceptable; the electric vehicle battery can be expected to last longer and perform mobility tasks more reliably. Finally, this method also demonstrates that the accuracy of estimating the state of charge (SOC) of a battery in an electric vehicle can be improved by utilising narrow learning layers.

Keywords*: *Battery, Corny Architecture, Electric Vehicle, State of Charge

I. INTRODUCTION

The increase in carbon emissions from vehicles running on Internal Combustion Engines (ICE) requires an alternative source of vehicle power. An Electric Vehicle (EV) is an impeccable solution to address this problem, thereby reducing environmental damage. In operating an Electric Vehicle, the most essential part of an EV is the energy storage devices [1]. The Energy storage devices and power supply used in high-power EV applications are batteries. The cutting-edge battery technology offers a wide range of advantages, including high energy density, low environmental pollution, and long cycle life. When attached to the load, the battery's efficiency depends on the chemical reactions that occur within it. The boom in the EV industry requires a significant demand for an effective battery management system. The battery management system includes battery monitoring. The temperature and other atmospheric variables most influence the performance parameters of the lead acid battery. Furthermore, battery monitoring is a vital task in most electric vehicles (EVs), as it affects the operation of the battery, personal safety, and the EV's lifespan. This monitoring helps maintain the battery's condition within the stated safety limit of operation. There were many methods available for monitoring the state of charge (SOC) of the battery so far. The management of the battery system using interleaved pulse charging is proposed to monitor battery management. It includes fast charging, battery ageing diagnosis, charge estimation, and balancing. This method used a single inductor, single input, dual output architecture [2]. It is shown in Fig. 1 below.

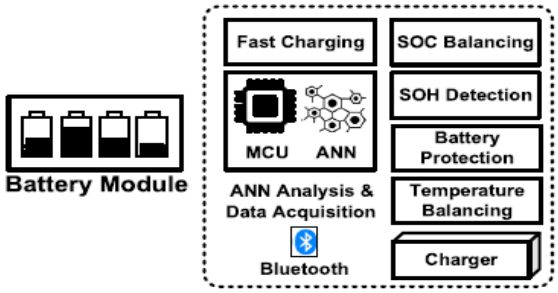


Fig. 1: The Battery Management System [2]

Interleaved pulse charging is designed to facilitate a fast charging process and delay battery ageing. Many surrounding factors, such as variations in cell temperatures, influence the ageing battery. This method also significantly suppresses the temperature fluctuation of the cell surface. It can facilitate charge balancing of the battery. This method utilises an artificial neural network to detect the state of health (SOH) of battery cells, thereby improving the accuracy of state of charge (SOC) estimation. The details of the architecture are as shown in Fig. 2.

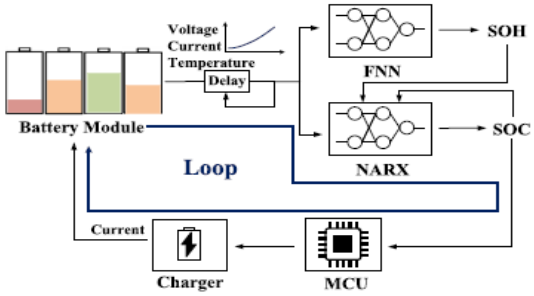


Fig. 2: The use of CNN to SOH Architecture [2]

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Fig. 2 shows an ANN with an input delay feedforward employed. The ANN process collected data from ADC on the MCU, including output voltage V_{cell}, charging current I_{module} and temperature of each battery cell V_{tem}. The sample data is sent to the ANN model to estimate the State of Charge (SOC) and State of Health (SOH). The other method for estimating battery state of charge is the multi-timescale estimator. This method provides accurate and robust estimation of the state of charge and state of energy in a battery management system [3]. The estimator can predict the state of charge (SOC) and state of energy (SOE) under dynamic operation and varying temperatures. The predictor utilises a dual H-infinity filter. The block diagram of this method is shown in Fig. 3 below.

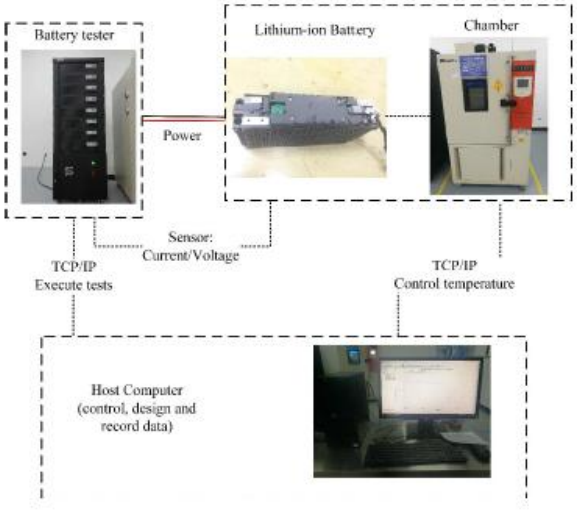


Fig. 3: The Block Diagram of Dual H Infinity Filter [3]

Fig. 3 shows a block diagram of the experiment to obtain the experimental data. The data consists of terminal voltage, current and OCV. The PC is used to control and store the data. A thermal chamber is used to set the battery's temperature. The battery tester performs strategy charging and recharging with a wide range of temperature levels. A TCP/IP module is used for data transmission. The result shows that the prediction of SOC and SOE has an error rate of approximately 1.5%. Further, other methods used a fractional Kalman filter to estimate the state of charge in battery management systems, especially for lithium-ion batteries [4]. The purpose of this method is to reflect the dynamic characteristics of the battery. Moreover, fractional orders were identified using a genetic algorithm. It is as shown in Fig. 4 below.

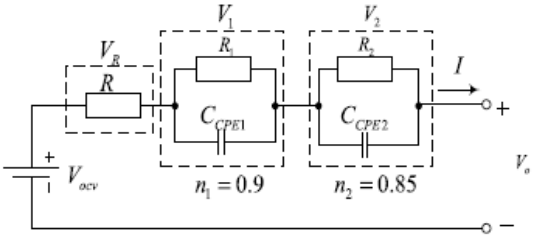


Fig. 4: The Fractional Kalman Filter [4]

Fig. 4 shows that the fractional Kalman filter circuit is based on a second-order RC model. Where V_o is the terminal battery voltage, V_{ocv} is the battery open-circuit voltage, V_R is the battery internal resistance voltage, and R is the internal resistance of the battery. The error rate of the terminal voltage is 0.014 V. This accuracy is considered reliable. The importance of a sound battery management system, especially in electric vehicles, lies in maintaining the battery's health and extending its long-term utility. Furthermore, the battery supports many vital roles. They are monitoring voltage and current, estimating charge and discharge, equalizing and protecting the battery, managing temperature and data, and so on[5]. The battery modelling, state estimation, and battery charging are crucial. The standard of electric vehicle is as shown in Fig. 5 below.

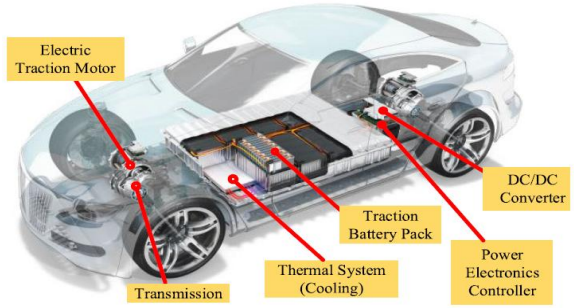


Fig. 5: The Standard of Electric Vehicle [5]

Fig. 5 shows the layout of the EV functional components standard. The design offers safety and reliability for EV driving and other uses. Another critical factor in extending the useful life of a battery is charge balancing among battery cells. The unbalanced charging among cells can reduce battery capacity, accelerate battery degradation, and even cause some safety hazards [6]. To handle this problem, the use of a battery charging equalisation system (BCE) is necessary. This method considers the battery as a series of cells. This circuit is referred to as a battery module. Module-level equalisers are connected to each pair of consecutive battery modules. These interconnected modules, featuring a new architecture, are known as the BCE system. As shown in Fig. 6.

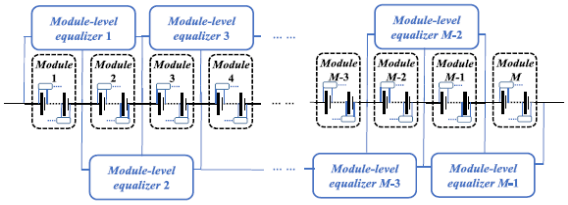


Fig. 6: The Architecture of BCE System [6]

The mathematical model of the BCE system is presented in Eqs. 1, 2, and 3. The charge transfer rate, kicl, is given according to Eq. 1.

$$K_i^{cl}(n) = \begin{cases} \mathrm{sgn}1(-\Delta_i^c(n-1), l_c) r_c, if, i-1 > (\Gamma_\beta^i) B, \\ 0, otherwise \end{cases} \quad (1)$$

$$K_i^{cr}(n) = \begin{cases} \mathrm{sgn}1(-\Delta_i^c(n-1), l_c) r_c, if, i-1 > (\Gamma_\beta^i) B, \\ 0, otherwise \end{cases} \quad (2)$$

Where

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$$\operatorname{sgn} l(u,v)=\begin{cases} -1, & \text{if } u<0, \\ 0, & \text{if } u=0, \\ 1\text{-}v, & \text{if } u>0, \end{cases} \tag{3}$$

$$\Delta_h^c(\text{n-1})=x_h(t_{n-1})-X_{h-1}(t_{n-1}) \tag{4}$$

Thus,

$$K_i^{ml}(n)=\begin{cases} \operatorname{sgn} l(-\Delta_i^c(n-1),l_m)r_m, if\ ,i-1>(\Gamma_\beta^i)-1>0, \\ 0, otherwise \end{cases} \tag{5}$$

$$k_i^{mr}(\text{n})=\begin{cases} \operatorname{sgn} l(\Delta_{1+b}^m(n-1),I_M)r_m, \text{ if } \Gamma\frac{i}{B}+1\geq M, \\ 0, \qquad \text{otherwise} \end{cases} \tag{6}$$

$$\Delta_h^m(n-1)=\sum_{j=\left(\left\lfloor\frac{h}{B}\right\rfloor-1\right)B+1}^{\left\lfloor\frac{h}{B}\right\rfloor B}X_j(t_{n-1})-\sum_{j=\left(\left\lfloor\frac{h}{B}\right\rfloor-2\right)B+1}^{\left(\left\lfloor\frac{h}{B}\right\rfloor-1\right)B}X_j(t_n-1) \tag{7}$$

Based on the above equations, the SOC changes from the i-th cell, that is,

$$k_i^{net}(n)=k_i^{cl}(n)+k_i^{cr}(n)+k_i^{ml}(n)+k_i^{mr}(n) \tag{8}$$

Eq.8 shows the cell SOC evolution during each working cycle.

$$X_i(t)=x_i(t_{n-1})+\frac{K_i^{net}(n)}{\tau}(t-t_{n-1}), \tag{9}$$
$$t\in(t_{n-1},t_n], \text{ n=1,2,3,...)}$$

Based on Eq. 9, the system can achieve charge equalisation. Then fully charge and discharge together without having overcharging and overdischarging. A joint state of charge (SOC) and state of available power (SOAP) method is used to estimate battery model parameters. The SOAP of the battery is estimated, then the joint SOC is performed [7]. The method achieved the joint SOC and SOAP, obtaining higher accuracy than the Adaptive Battery State Estimator (ABSE) method. The ABSE algorithm errors originate from the identification of internal resistance (Rp). Where the identified values are much higher than the actual values when the battery is charged or discharged at a high current. The circuit of the estimation model is as shown in Fig. 7 below.

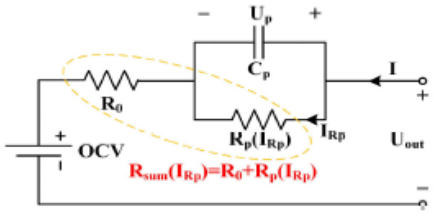


Fig. 7: The Circuit Model of Joint SOC and SOAP Method [7]

The testing of joint SOC and SOAP is performed according to the schematic diagram below. It is shown in Fig. 8.

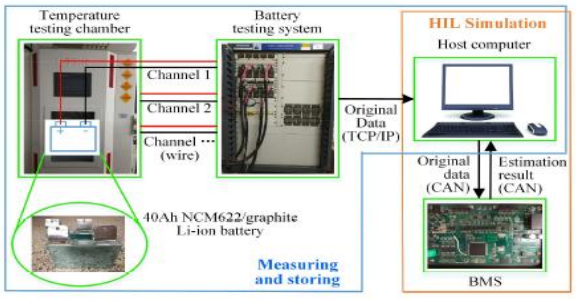


Fig. 8: Schematic Diagram of Battery Test Bench [7]

This estimation was more accurate compared to the ABSE method. The other method for estimating the state of charge (SOC) of the battery involved using a typical extended Kalman filter. This method is an improvement of the Kalman filter algorithm or particle filtering algorithm. This method generates an error of about 5% to 10%, depending on battery characteristics [8]. The typical extended Kalman filter has a flow chart with four phases. The flow chart is as shown in Fig. 9 below.

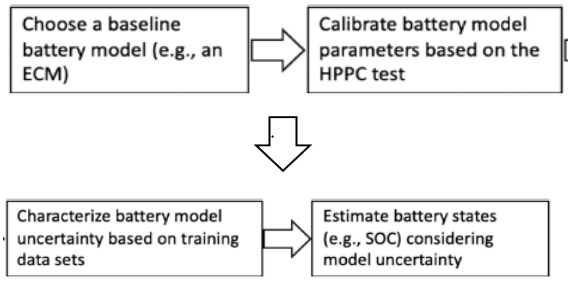


Fig. 9: Flow Chart of Joint SOC [8]

The improvement can be achieved using a Gaussian filter and a polynomial regression model. This technique can examine the effect of SOC and SOAP methods on a typical battery circuit model. The increase in large-scale electric vehicles causes fluctuations in charging and discharging voltage. A method is needed to stabilise the battery's voltage. The improvement of battery lifecycles can be reached. The SOC needs to work at a higher level. It is necessary to monitor the frequency of the charging/discharging voltage[9]. The details of the optimised SOC have the scheme as shown in Fig. 10.

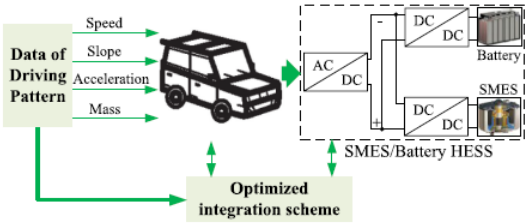


Fig. 10: The Optimised SOC [9]

The development of the SOC scheme utilised the unscented Kalman filter (UKF) method, and an extended Kalman filter was employed to estimate the internal resistance of the battery. The process can estimate system parameters and the battery's state of charge. The test bench of the process is as shown in Fig. 11.

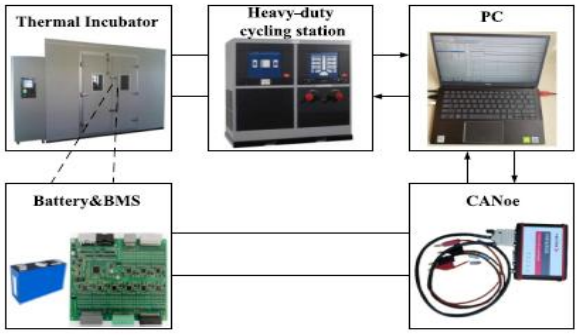


Fig. 11: The Test Bench of UKF Model [9]

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The monitoring and estimation of reliance batteries were urgent, especially for those with lower energy and power density. This kind of battery is prone to ageing and performance degradation over time, and restricts their mainstream adoption [10]. The method used for handling the above task is Digital Twin (DT)[10]. This method was applied to the difficulty of onboard computation for incremental SOC and SOH. The schematic of DT is as shown in Fig. 12.

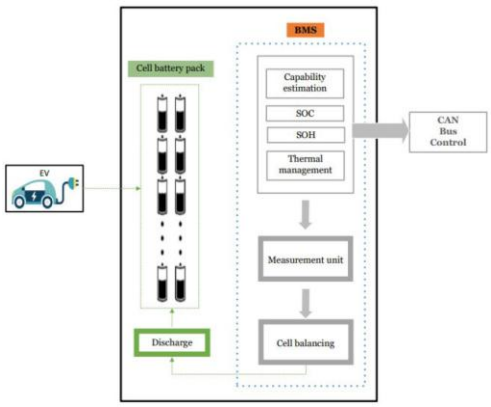


Fig. 12: Schematic diagram of DT [10]

The electrical circuit model of the battery is as shown in Fig. 13.

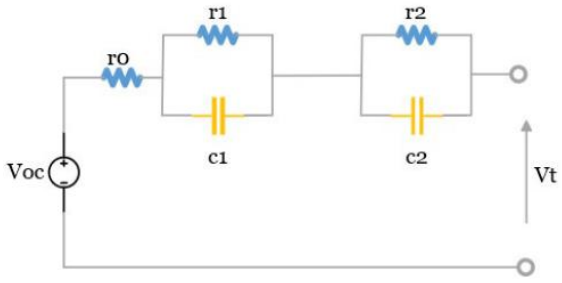


Fig. 13: The Circuit Model of ECM Structure [10]

To predict the SOH and SOC, the mathematical formulation is according to Eq.10

$$(SOC)_{\tau} = 1 - \int_0^T \delta \frac{i}{c_n} dt \qquad (10)$$

This method helps manage batteries and complete life cycle statistics, as well as the upgrade path for the batteries.

II. DEEP LEARNING ON BMS

Deep learning is derived from conventional neural networks but outperforms its predecessor [11]. Deep learning employs transformations and graph techniques simultaneously to build multilayer training models. There were several methods for acquiring deep knowledge within the BMS system. The use of deep learning based on a transformer model trained with self-supervised learning (SSL). The SSL framework generated estimation with the accuracy root mean square (RMS) of 0.9 [12]. A neural network with an Elman architecture is employed to estimate the State of Charge (SOC) and State of Health (SOH) of a lithium-ion cell. The multi-objective optimisation approach based on Particle Swarm Optimisation (PSO) for training to achieve a lower root mean square (RMS) error. The number of hidden layers, activation functions, bias values, and the weights of inputs and outputs influences the optimisation of neural network characteristics [13]. The details of the Elman architecture are as shown in Fig. 14.

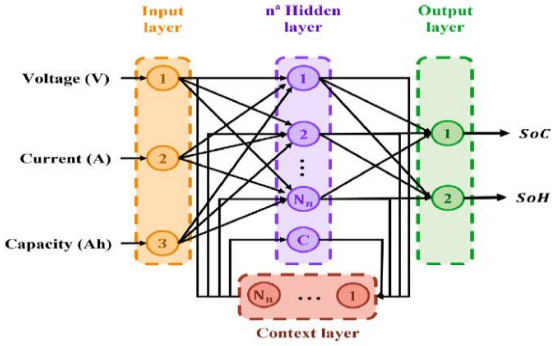


Fig 14. The Elman Architecture [13]

The combination of a Kalman filter and deep learning forms a hybrid method for estimating the state of charge (SOC) of Li-ion batteries. The first CNN was combined with different versions of RNN, including LSTM, to obtain SOC [14]. This method can reduce the computational cost. State of the art in determining SOC and SOH of battery is critical because it can optimise the battery energy strategy [15]. This method can estimate the non-linear and time-varying charging and discharging characteristics of a battery. The other method utilises deep learning with optimisation using the Evolutionary Mating Algorithm (EMA). The EMA algorithm is used for optimization of DL parameters to estimate the SOC of a battery for an electric vehicle [16]. It can be employed as a proficient technique to accurately estimate the state of charge (SOC) of electric vehicle batteries.

III. METHOD

The proposed method uses deep learning with narrow learning layers. It is called corny learning architecture. This corny model contains a sequence input, a fully connected layer, an activation function (tanh function), a fully connected layer, a ReLu layer, a fully connected layer, a ReLu layer, and a regression output layer. This typical model enables the model to adapt to varied data. The learnerable layers respond quickly to every change in the fluctuating input. The use of some fully connected layers generates the function form accurately and faster. The details of the filter size architecture are as shown in Table 1.

Table 1. The Filter Size of Each Layer of the Convolutional Model

sequenceinput Sequence input with 5 dimensions	5(C) × 1(B) × 1(T)
fc_1 55 fully connected layer	55(C) × 1(B) × 1(T)
layer Hyperbolic tangent	55(C) × 1(B) × 1(T)
fc_2 55 fully connected layer	55(C) × 1(B) × 1(T)
leakyrelu Leaky ReLU with scale 0.3	55(C) × 1(B) × 1(T)
fc_3 1 fully connected layer	1(C) × 1(B) × 1(T)
clippedrelu Clipped ReLU with ceiling 1	1(C) × 1(B) × 1(T)
regressionoutput mean-squared-error	1(C) × 1(B) × 1(T)

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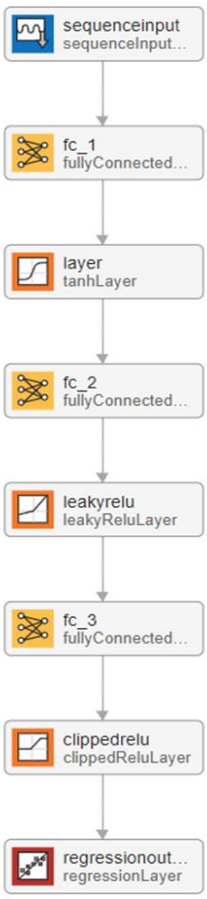


Fig. 15: The Corny Architecture Model

Figure 15 shows the sequence of the proposed method. It starts with an input layer, a fully connected layer, an activation layer, a second fully connected layer, a second activation layer, a third fully connected layer, a third activation layer, and a regression layer. This architecture does not include convolutional layers, pooling layers, and batch normalisation layers. The regression layer predicts the values of the target. The architecture has two-dimensional filters. It lacks a depth dimension. The prediction value is a number. The regression layer performs this task. The data is taken from the Lithium-ion dataset. It contains voltage, current, temperature and capacity of the battery[17]. Input parameters influence the model generated from the training data. This is illustrated in Fig. 16.

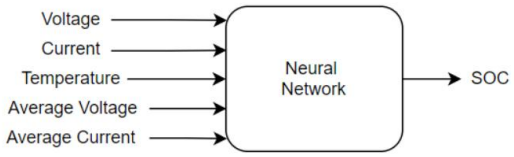


Fig. 16: The Model Architecture of the Corny Model

The characteristic battery is varied, which refers to the influence of external factors. Due to the nonlinear temperature, health, and state-of-charge (SOC)-dependent behaviour of Li-ion batteries, this method enriches previous deep learning methods with a simpler architecture. A more complete architecture might lead to an overfitting problem and higher costs. Additionally, training is also complicated since the dataset setting and filter bank size are crucial in deep learning. This method prevents the complexity and possible overfitting problem that can occur as long as the training process takes place. The training sample trains a neural network to predict the state of charge of a Li-ion battery, given time series data representing various features of the battery, such as voltage, current, temperature, and average voltage and current (over the last 500 seconds). It contains a sequence of data collected while the battery-powered electric vehicle was driven during a driving cycle with an external temperature of 25 degrees Celsius. The test data includes four sequences of data collected during driving cycles at four different temperatures. They are 0 degree C, 10 degree C, 25 degree C, 20 degree C and 40 degree C. Furthermore, the training hyperparameters are set to train for 1200 epochs with mini-batches of size 1 using the "Adam" solver. To prevent the gradients from exploding, set the gradient threshold to 1. Specify an initial learning rate of 0.01, a learning rate drop period of 400 and a learning rate drop factor of 0.1. Specify a validation frequency of 30. The initial learning rate of 0.01 and the learning rate drop factor of 0.1 together minimize the validation error.

IV. RESULT

The training process is shown in Fig. 17, with its progress.

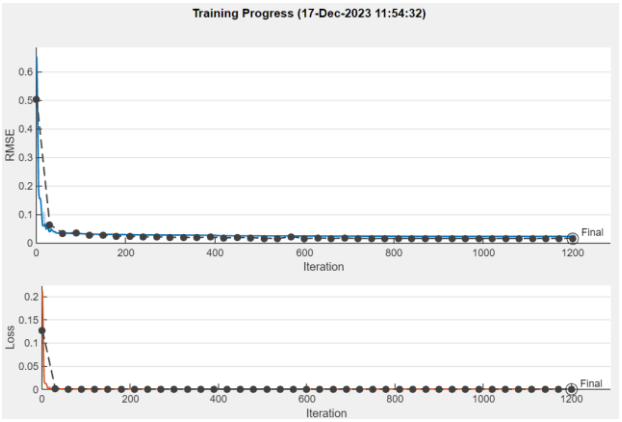


Fig. 17: The Progress of the Training Process

Based on the previous hyperparameters and input data, the training process generates an error rate of less than 0.02. Further, the losses obtained are very close to zero and might be zero. It means that every input data offered an accuracy of 100% at 1200 iterations. The estimation of the soil organic carbon (SOC) is accurate. As shown in Fig. 17, the training process achieves a low RMSE very quickly. It requires fewer than 50 iterations. At the 30th iteration, it reaches an MRSE of about 0.05%. Then, at the 60th iteration, it reaches an MRSE of less than 0.025, and it remains steady for the subsequent iterations, at about 0.001 of the MRSE index.

Furthermore, the same MRSE characteristic also occurs in the progress of the loss index. The training process, with 1,200 iterations, achieves a rapid speed in reaching a low loss index. Fortunately, it can get the lowest value or zero. The fast movement of the decrement loss index occurs at about the 30th iteration. For the remaining iterations, the loss index exists at zero values. To reassure that the generated model is accurate, testing data from various environmental temperatures are employed with the model. The result is as shown in Fig. 18 below.

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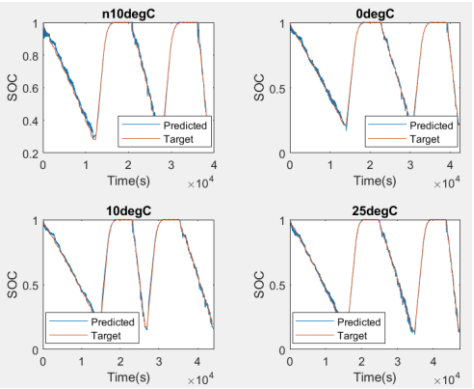


Fig. 18: The Estimation of SOC Battery at Various Temperatures

Figure 18 shows a graph with predicted and target SOC at different temperatures. The blue graphs represent estimated values, while the red ones represent target values. The highest accuracy was achieved when the blue and red graphs were merged. Nevertheless, the accuracy is poor when a curved distance separates the red and blue graphs.

When different methods are used to assess the measurement error, they include the root mean square error (RMSE) and the maximum absolute error (MAX) at various temperatures. The lower the value index at the same plot temperature, the accurate the estimation is. The proposed method has RMSE and MMAX indexes as shown in Fig. 19.

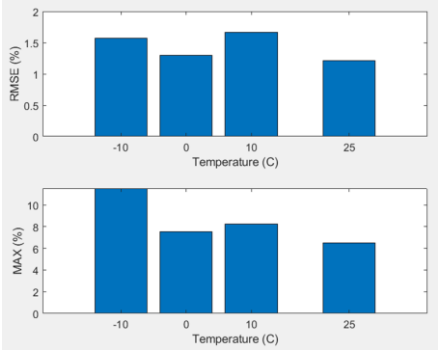


Fig. 19: The RMSE and MAX Indexes of the Corny Model

Fig. 19 shows that the RMSE index of the proposed method is very low. It has a range of 0 to 1.5. The plot with the lower index represents measurements performed at 0°C and 25 °C. Nevertheless, the temperatures of 10 °C and -10 °C have the highest root mean square error (RMSE). This nature also happens to be the maximum absolute error index.

Finally, the proposed method, the Corny model, can accurately estimate the HOC of electric vehicles.

V. CONCLUSION

The measurement of SOC and HOC in a battery is crucial, particularly during the operation of an electric vehicle. It determines the battery's long life and reliability.

In this paper, a SOC and HOC estimation method for lithium-ion batteries is proposed, based on a typical deep learning architecture. Firstly, the data contains parameters such as temperature, capacity, voltage, and current of the Li-ion battery, which are labelled. Then, based on those properties, the design model architecture is performed. The architecture network is very narrow to prevent high cost and the over-fitting problem. The Corney model can estimate the state of charge (SOC) of Li-ion batteries with very high accuracy. The RMSE and MAX indexes are very low. Thus, the temperature also influences the MRSE and MAX indexes of the Li-Ion battery. This accuracy and lower cost index are the two advantages of the proposed method. Additionally, the design is straightforward and can be easily understood by individuals with no prior experience in deep learning programming.

Acknowledgment.

DECLARATION STATEMENT

Authors are required to include a declaration of accountability in the article, including review-type articles, that stipulates the involvement of each author. The level of detail differs; some subjects yield articles that consist of isolated efforts that can be easily detailed, while other areas function as group efforts at all stages. It should be after the conclusion and before the references.

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